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Cytoskeletal rearrangements are accompanied by marked changes in local mechanical properties that can be measured using a new technique called live cell multiple particle tracking. It has been well established that Rho GTPases play a central role in cytoskeletal assembly, but the downstream mechanical response of the cytoskeleton remains unknown. The goal of this proposal is to characterize the mechanical response of the cytoskeleton to endogenous GTPase activation (e.g., growth factors, microgravity) and to elucidate the observations made during the Russian Biocosmos 10 space mission where adherent cells demonstrated dramatic changes in cytoskeletal organization and related contacts to the substratum. The proposed research will require limited use of the reduced hydrodynamic shear bioreactor in the biotechnology cell science program at the Johnson Space Center. Cell cultures will be maintained in the bioreactor for a period of time not to exceed 1 week in order to simulate the microgravity of space. The cultures will then be lysed and analyzed at the Johnson Space Center with materials prepared at Johns Hopkins University. AIM 1. We shall develop a new high-throughput cell-mechanics method, which involves a novel nanosphere injection. The time-dependent

Task Description:	coordinates of injected nanospheres will be video-tracked with high spatial resolution; the coordinates will be transformed into mean squared displacements and finally into local frequency-dependent viscoelastic moduli. Using this novel cytomechanics method, called multiple particle tracking microrheology (MPTM), we shall map the micromechanical properties of resting (i.e. quiescent) Swiss 3T3 cells. We shall quantify the ensemble-averaged, frequency-dependent viscoelastic moduli of these cells as well as their spatial distributions within cells (subcellular mapping of cytomechanics) and assess cell-to-cell variations in culture. AIM 2. We hypothesize that Rho-mediated cytoskeleton re-organization transiently changes the micromechanical properties of Swiss 3T3 cells. Using live-cell MPTM, we shall map the time-dependent micromechanical properties of the cytoplasm of Swiss 3T3 cells upon activation of the small GTPase Rho by lysophosphatidic acid (LPA). Using a biochemical pull-down assay, we shall directly cross-relate these measurements with the time-dependent level of active Rho. Rho activation and associated cell mechanics will also be assessed as a function of LPA concentration. AIM 3. We hypothesize that the activation of the small GTPases Rac and Cdc42 regulate the mechanical properties of cells in a spatially heterogeneous manner. To test this hypothesis, we shall use MPTM to map the time-dependent micromechanical properties of Swiss 3T3 cells upon the individual activation of Rac and Cdc42 with platelet-derived growth factors (PDGF) and Bradykinin, respectively. We will then extend this analysis to cells exhibiting polarized movements, which is known to require local activation of Rac and Cdc42, to examine the local micromechanics of cell locomotion. These MPTM measurements will be complemented by biochemical characterization of levels of active Rac and Cdc42. AIM 4. We hypothesize that microgravity conditions effects activate small GTPases. Using a biochemical pull-down assay, we shall determine the levels of active GTPases in cells subjected to simulated microgravitational conditions compared to normal gravity. Cells will be cultured in the Johnson Space Center's reduced hydrodynamic shear bioreactor. To mimic cell's response during a wound-healing event, we shall monitor the levels of active Rho, Rac, and Cdc42 as a function of time upon lysophosphatidic acid (LPA) and PDGF treatment. The cytoskeleton is a highly dynamic structure that provides cells with strength and rigidity. It is well established that cytoskeletal rearrangements are closely correlated with many cellular processes such as cell locomotion, cell shape changes during mitosis, contractile activities such as muscle contraction or the separation of daughter cells by the contractile ring during cytokinesis, cell-cell and cell-substrate interactions together with adhesion molecules, and transmembrane signaling, endocytosis, and secretion. However, the resulting changes in the mechanical properties of the cell are not well-understood. Local mechanical properties of cytoplasm have been implicated in the control of cell shape and the direction of locomotion, and there is now evidence that oncogenic human epithelial cells possess mechanical properties that are strikingly different from those of normal human epithelial cells. Current imaging methods cannot readily produce quantitative measurements of the structural and mechanical microheterogeneity of the cytoskeleton of living cells. Electron microscopy (EM) combined with immunocytochemistry has been successfully used to obtain detailed insight into the distribution and molecular interactions of macromolecular assemblies in situ. Similarly, cryo-electron microscopy techniques (e.g. rapid-freeze deep-etching) preserve the three-dimensional organization of the cytoskeleton. However, EM has not been extended to quantify the degree of spatial heterogeneity of networks, does not probe physical properties (e.g., cytoplasmic viscosity, etc.), and does not allow live-cell investigations. This last limitation can be alleviated partly by using correlative light and electron microscopy, which combines the live-cell capabilities of light microscopy and the superior spatial resolution of EM. Confocal laser scanning microscopy can visualize the three-dimensional organization of cytoskeletal elements in situ, but makes use of fluorescent dyes that can affect intermolecular interactions and does not measure physical properties. The method of photon spectroscopy in highly scattering media probes the dynamics of tissues and cytoskeletal structures in solution with superior temporal resolution, but sacrifices rich local structural information. Magnetic tweezers were recently used to probe the frequency-dependent viscoelastic moduli of cells. However extremely large (magnetic) beads were used (1.3 μm), the interactions between the bead subcellular structures were not characterized, and the surface area of contact between bead and cell is ill-defined; moreover this approach is ill-suited to probe the local cellular response to an extracellular stimulus. Atomic force microscopy has been used to monitor the mechanical properties of cells, but it does not provide frequency-dependent viscoelastic moduli and can also be too slow to monitor rapidly changing mechanical properties. Obtaining frequency dependence is essential as different rates of deformation may lead to completely different viscoelastic behavior. By monitoring the diffusion of small, inert tracer particles via fluorescence microscopy, Ragsdale et al. probed the micromechanical properties of fibroblasts. However, many assumptions were made to compute the cytoplasmic elasticity and no global cellular response was obtained. It is the aim of this proposal to use a new technique called multiple particle tracking microrheology to investigate the local micromechanics of cell locomotion and to monitor the micromechanical response of fibroblasts to the activation of specific members of the Rho family of small GTPases. Rho proteins cycle between an active GTP-bound state and an inactive GDP-bound state, and translate extracellular and intracellular signals into coordinated cytoskeletal events. Of the 18 known Rho proteins, Cdc42, Rac, and Rho are the most thoroughly characterized 15, and each of these proteins contributes to cell motility by stimulating rearrangements in the actin cytoskeleton 16. Currently, the roles of Rho proteins in actin filament and network dynamics are fairly well understood at the biochemical level, however the mechanical mechanisms by which they control cell locomotion is not well understood. We propose to study the mechanics of cell locomotion in concert with the mechanics of Rho GTPase-mediated cytoskeletal organization to advance the understanding of cell motility. We shall also determine how small GTPases are activated in cultured cells in microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The focus of this NASA-funded research is primarily to investigate the effect of microgravity on cell bahavior, in particular cell mechanical response. This research has resulted in the development of an extremely versatile method, called multiple-particle tracking microrheology, which allows an inside-view of cell mechanics, as opposed to the more common back box experiments currently done. This approach is now being used by us and others to investigate the micromechanical response of various chemical and physical stimuli on Earth. Given the remarkable power of this approach, we foresee that our method will become the standard approach in cell mechancs within the next few year.

Task Progress:	Cytoskeletal rearrangements are accompanied by marked changes in local mechanical properties that can be measured using a new technique called live cell multiple particle tracking. It has been well established that Rho GTPases play a central role in cytoskeletal assembly, but the downstream mechanical response of the cytoskeleton remains unknown. The goal of this proposal is to characterize the mechanical response of the cytoskeleton to endogenous GTPase activation (e.g., growth factors, microgravity) and to elucidate the observations made during the Russian Biocosmos 10 space mission where adherent cells demonstrated dramatic changes in cytoskeletal organization and related contacts to the substratum. The proposed research will require limited use of the reduced hydrodynamic shear bioreactor in the biotechnology cell science program at the Johnson Space Center. Cell cultures will be maintained in the bioreactor for a period of time not to exceed 1 week in order to simulate the microgravity of space. The cultures will then be lysed and analyzed at the Johnson Space Center with materials prepared at Johns Hopkins University. During the last year and with funding from NASA, we have achieved extraordinary progress on proposed work in four interrelated areas of research: With NASA funding (acknowledged in the papers), we have published 12 papers and have achieved the following tasks: (1) Cell polarization and nucleus positioning. Intracellular migration of the nucleus is required for proper tissue growth and development as well as nucleus positioning in eukaryotic cells. The molecular regulators of nucleus movement in interphase fibroblasts have not been identified. Here, we show that the nucleus of Swiss 3T3 fibroblasts subject to flow shear stresses display rigid-body motion, which depends on the integrity of microtubules, not that of F-actin. Through inactivation of major members of Rho GTPases, well-known mediators of cytoskeleton reorganization, we find that Cdc42, not RhoA and Rac1, controls the degree of rotation and extent of translational movements of the nucleus. Shear stimulus in Swiss 3T3 fibroblasts causes MTOC repositioning and cell polarization in the direction of flow, a process controlled by the Cdc42-Par6-PKC? pathway, not RhoA nor Rac1. These results establish Cdc42 and microtubules as major molecular regulators of nucleus movement and MTOC orientation in Swiss 3T3 fibroblasts. (2) Micromechanics of migrating fibroblasts. Cell migration is a highly coordinated process that occurs through the translation of biochemical signals into specific biomechanical events. The biochemical and structural properties of the proteins involved in cell motility, as well as their subcellular localization, have been studied extensively. However, how these proteins work in concert to generate the mechanical properties required to produce global motility is not well understood. Using intracellular microrheology and a fibroblast scratch-wound assay, we show that cytoskeleton reorganization produced by motility results in mechanical stiffening of both the leading lamella and the perinuclear region of motile cells. This effect is significantly more pronounced in the leading edge suggesting that the mechanical properties of migrating fibroblasts are spatially coordinated. Disruption of the microtubule network using nocodazole results in the arrest of cell migration and a loss of subcellular mechanical polarization, however, the overall mechanical properties of the cell remain mostly unchanged. Furthurmore, we find that activation of Rac and Cdc42 in quiescent fibroblasts elicits mechanical behavior similar to that of migrating cells. We conclude that a differential distribution of subcellular mechanical microenvironments is essential for directed cell migration and is coordinated through microtubules. (3) Cell mechanics – synergistic effect of actin-crosslinking proteins. Actin-crosslinking proteins organize actin filaments into dynamic and complex subcellular scaffolds that orchestrate important mechanical functions, including cell motility and adhesion. Recent mutation studies have shown that individual crosslinking proteins often play seemingly non-essential roles, leading to the hypothesis that they have considerable redundancy in function. We report live-cell, in vitro, and theoretical studies testing the mechanical role of the two ubiquitous actin-crosslinking proteins, a-actinin and fascin, which co-localize to stress fibers and the basis of filopodia. Using live-cell particle tracking microrheology, we show that the addition of a-actinin and fascin elicits a cell mechanical response that is significantly greater than originated by a-actinin or fascin alone. These live-cell measurements are supported by quantitative rheological measurements with reconstituted actin filament networks containing pure proteins that show that a-actinin and fascin can work in concert to generate enhanced cell stiffness. Computational simulations using finite element modeling qualitatively reproduce and explain the functional synergy of a-actinin and fascin. These findings highlight the cooperative activity of fascin and a-actinin and provide a strong rationale that an evolutionary advantage might be conferred by the cooperative action of multiple actin-crosslinking proteins with overlapping but non-identical biochemical properties. Thus the combination of structural proteins with similar function can provide the cell with unique properties that are required for biologically optimal responses. (4) Micromechanics of reconstituted nucleus lamina. Lamin B1 filaments organize into a thin dense meshwork underlying the nucleoplasmic side of the nuclear envelope. Recent experiments in vivo suggest that lamin B1 plays a key structural role in the nuclear envelope, but the intrinsic mechanical properties of lamin B1 networks remain unknown. To assess the potential mechanical contribution of lamin B1 in maintaining the integrity and providing structural support to the nucleus, we measured the micro-mechanical properties and examined the ultra-structural distribution of lamin B1 networks in vitro using particle tracking methods and differential interference contrast (DIC) microscopy. We exploit various surface chemistries of the probe microspheres (carboxylated, polyethylene glycol-coated, and amine-modified) to differentiate lamin-rich from lamin-poor regions and to rigorously extract local viscoelastic moduli from the mean squared displacements of non-interacting particles. Our results show that human lamin B1 can, even in the absence of auxiliary proteins, form stiff and yet extremely porous networks that are well suited to provide structural strength to the nuclear lamina. Combining DIC microscopy and particle tracking allows for the first time to relate directly the local organization of a material to its local mechanical properties, a general methodology that can be extended to living cells.

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